

Compound 1

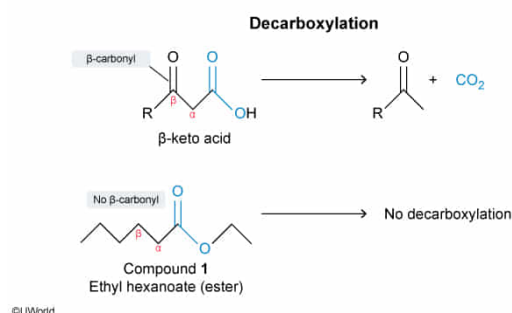
Compound 1, a carboxylic acid derivative, is shown above. The carboxylic acid it is derived from cannot undergo decarboxylation because Compound 1:

- ☒ A. does not contain a β -carbonyl. [46%]
- ☐ B. contains a conjugated ester. [26%]
- ☐ C. contains a β -diketone. [4%]
- ☐ D. does not contain an acidic proton. [22%]

Correct

47% answered correctly

Explanation:



Decarboxylation is a reaction that **removes** a **carboxyl** group from a **carboxylic acid** with a **β -carbonyl**, releasing the carboxyl group as **CO₂ gas**. An **ester** with a β -carbonyl can also **undergo decarboxylation** if the ester is first **hydrolyzed** to a carboxylic acid. Once the carboxylic acid is formed, decarboxylation occurs through a **cyclic transition state** that incorporates both carbonyls, releases CO₂, and forms an **enol**. The enol tautomerizes to the keto form, or the α -carbon can be **alkylated**. A ketone is formed in both cases.

Compound 1, a carboxylic acid **derivative**, is the ester ethyl hexanoate, which does not contain a β -carbonyl. Therefore, the carboxylic acid it is derived from (ie, hexanoic acid) cannot undergo decarboxylation.

(Choice B) The carboxylic acid derivative is an ester, but it is not **conjugated** with another functional group. In addition, conjugation is not required for decarboxylation.

(Choice C) A β -carbonyl is required for decarboxylation, but a carboxyl group is also necessary. A β -diketone, a functional group made up of two carbonyls separated by a CH_2 , cannot undergo decarboxylation because it does not contain a carboxyl group.

(Choice D) Esters do contain an acidic proton (α -hydrogen), but acidic protons do not take part in the decarboxylation mechanism.

Educational objective:

Decarboxylation is a reaction that removes a carboxyl group from a carboxylic acid with a β -carbonyl, releasing the carboxyl group as CO_2 gas. A β -carbonyl is necessary for decarboxylation because a cyclic transition state incorporating both carbonyls is formed. Esters with a β -carbonyl can also undergo decarboxylation if they are hydrolyzed to a carboxylic acid first.

Time spent:0 sec

Subject:

Organic Chemistry

QID:402584

System:

Functional Groups and Their Reactions